

SUPPLY CHAIN MANAGEMENT ANALYSIS REPORT

Interactive Power BI Dashboard for Cross-Functional Performance Monitoring

Prepared For: Portfolio Project / Placement
Presentation / MCA Documentation
Platform Focus: Microsoft Power BI Enterprise
Architecture

Analysis Scope: Multi-Category Operations
(Furniture, Electronics, Clothing, Accessories)
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Project Summary

The **Supply Chain Management Analysis Dashboard** is an interactive Business Intelligence solution developed in Microsoft Power BI to evaluate cross-functional operational performance. The platform integrates distinct organizational data flows—warehouse logistics, inventory optimization, multi-regional transportation costs, annual sales cycles, and order fulfillment lines—into a single interactive monitoring ecosystem.

The core objective of this deployment is to translate raw business metrics into actionable operational insights. By exploiting robust dynamic cross-filtering and parameterized parameters, management can systematically slice, dice, and query metrics across all geographical environments and core product fields instantly.

Project Objectives & Dashboard Capabilities

Core Strategic Objectives

- Monitor volumetric warehouse utilization and infrastructure space layout.
- Evaluate stock cycle speeds via localized product turnover speeds.
- Benchmark shipping expenses across target geographical divisions.

Interactive Elements

- **Dynamic KPIs:** Core metric cards adjusting instantaneously to filters.
- **Regional & Category Slicers:** Dynamic multi-select drill-downs.
- **Cross-Filtering Framework:** Selection inside any chart filters the underlying visual pipeline model automatically.

- Assess supplier lead-time velocity to avoid structural fulfillment delays.

Key Performance Indicators (KPIs) Explained

WAREHOUSE UTILIZATION

32% - 35%

Measures structural warehouse square footage utilization.
Prevents overhead capacity waste.

DAYS SALES OF INVENTORY

14.8 - 16.3d

Average timeframe required to turn physical stock into realized sales revenue lines.

INVENTORY TURNOVER

22.1 - 23.8x

The annual velocity or number of full replacement cycles observed for core stock.

Dashboard Analysis by Product Category

Below is the detailed granular breakdown of performance parameters surfaced when dynamically querying individual inventory sectors via the dashboard's product slicers:

1. Furniture Category Segment

Utilization:

32.39%

DSI:

16.38 Days

Turnover:

22.17

Sales Trend:

Rapid volume climb peaking at ~14.6K units (2022), followed by structural softening through 2023 and 2024, prompting demand management strategies.

Logistics & Costs:

The West region represents severe freight spend outlays, directly tied to the high volume/mass characteristics of furniture items.

Fulfillment Status:

60 Orders Fulfilled | 21 Pending | 2 Cancelled. Highly stable fulfillment flow. Supplier lead time stands at 15.33 days.

2. Electronics Category Segment

Utilization:

34.42%

DSI:

14.95 Days

Turnover:

23.82

Sales Trend:

The primary organizational growth driver. Exploded from 4.0K units (2020) up to 18.0K units (2024), signaling scaling retail market penetration.

Logistics & Costs:

High total transportation footprint, dictated by fragile handling restrictions, security considerations, and high individual component valuation.

Fulfillment Status:

59 Orders Fulfilled | 24 Pending | 9 Cancelled. Note: Elevated cancellations indicate last-mile friction or spot localized stockouts. Lead time is 16.34 days.

3. Clothing Category Segment

Utilization: 35.01%

DSI: 14.89 Days

Turnover: 23.57

Sales Trend:	Steady initial growth peaking cleanly at 16.6K units in 2023, followed by a slight cyclical contraction down to 12.9K units in 2024.
Logistics & Costs:	Highly optimized regional cost profiles. Lower spatial parameters reduce physical storage constraints.
Fulfillment Status:	54 Orders Fulfilled 18 Pending 6 Cancelled. Balanced pipeline metrics. Lead time centers around 15.58 days.

4. Accessories Category Segment

Utilization: 35.12%

DSI: 15.83 Days

Turnover: 23.69

Sales Trend:	Volatile but resilient demand curve. Moved from 2.3K (2020) to 11.2K (2022), flattening slightly before climbing again to 11.6K units in 2024.
Logistics & Costs:	Lowest absolute freight expenses across categories due to compact product geometry and parcel delivery consolidation.
Fulfillment Status:	38 Orders Fulfilled 18 Pending 8 Cancelled. Shows reasonable baseline stability. Lead time is 15.81 days.

Comparative Category Analytics

The following cross-tabulation isolates operational matrices across product divisions to highlight outlier scenarios and optimize ongoing asset replenishment frameworks:

Product Category	Warehouse Utilization (%)	Days Sales of Inventory (DSI)	Inventory Turnover Ratio	Primary Operations Matrix Analysis
Furniture	32.39%	16.38 Days	22.17	Extreme freight cost sensitivity; volume tapering post-2022.
Electronics	34.42%	14.95 Days	23.82	Highest sales volume & turnover velocity; needs order cancellation review.
Clothing	35.01%	14.89 Days	23.57	Stable operations; seasonal volatility requires predictive demand safety stock.
Accessories	35.12%	15.83 Days	23.69	Low distribution space/freight cost impact; demand exhibits volatility.

Synthesized Business Insights & Recommendations

1. Operational Infrastructure Strategy

Across all categories, warehouse volumetric utilization tightly centers around **32% to 35%**. This signals considerable physical footprint safety headroom. Instead of maintaining empty space overhead liabilities, the company should aggressively look to scale its high-margin electronics portfolio or introduce new product lines to maximize return on asset layouts.

2. Transportation & Freight Rationalization

The West region represents an acute cost strain for the Furniture sector. Because these products are heavy and bulky, logistics teams must optimize consolidation strategies, renegotiate local zone freight agreements, or run warehouse routing algorithms to shift distribution points closer to high-density consumer centers.

3. Service Level & Order Precision Audits

While total order fulfillment remains high across the board, the **Electronics division exhibits an anomalous cancellation count (9 orders)**. Given that electronics represent the primary growth category (surging to 18K units in 2024), these pipeline failures erode customer retention and imply stockouts. Immediate priority must be assigned to fixing core supplier lag times (currently tracking at a high 16.34 days).

Strategic Summary Note for Stakeholders: Maximizing margins requires a dual approach: exploiting empty space overhead capabilities to hold inventory for high-velocity Electronics, while restricting capital allocations to aging or slow-moving Furniture segments.

Conclusion

The **Supply Chain Management Analysis Dashboard** demonstrates the transformative power of interactive Business Intelligence in managing complex global supply networks. By connecting disparate data sources—warehouse capacity benchmarks, operational pipeline turnover rates, real-time regional freight visibility, and fulfillment accuracy metrics—into an integrated Power BI canvas, the project gives management absolute operational transparency.

The empirical insights derived from this model clearly designate Electronics as the business's primary growth sector, while flagging Furniture as a candidate for intensive logistics margin protection. Ultimately, this business intelligence deployment transitions an organization from reactive crisis troubleshooting to a proactive, data-validated, and market-responsive supply chain champion.